

BEC303 Network Analysis — Assignment 5 —

Characterising and terminating two-port networks

Module 5: Two-Port Networks

Field	Value
Maximum marks	20
Weightage	10% of CIE
Issue date	<<TBD>>
Due date	<<TBD>>
Items	10 problems + 3 theory questions
CO / Bloom coverage	CO5: 13 · L1–L2 0, L3–L4 6, L5–L6 7

Objective

Develop the ability to characterise a network as a black box, choose the parameter set appropriate to the question, and convert a characterisation into terminated performance figures.

Learning outcomes

- Student will be able to determine z , y , h and ABCD parameters for a given network and state their terminations and units.
- Student will be able to convert between parameter sets and identify when a target set does not exist.
- Student will be able to test reciprocity and symmetry and use them as arithmetic checks.
- Student will be able to compute input impedance, output impedance and gain for a terminated two-port.

Part A — Problems (17 marks)

P1. For a T network of your own choice, determine all four z parameters from first principles, stating the termination used for each, and verify reciprocity and symmetry. [C05 | L3 | 2 marks]

P2. For the same network, determine all four y parameters directly from short-circuit conditions, then confirm they match the matrix inverse of your z result. [C05 | L4 | 2 marks]

P3. Derive the π equivalent from your y parameters and verify by analysing the π network that it reproduces the same four parameters. [C05 | L4 | 2 marks]

P4. Determine the ABCD parameters of the same network and verify $AD - BC = 1$. Explain what a failure of that check would have indicated. [C05 | L4 | 2 marks]

P5. Construct a network that has no z representation, demonstrate the failure explicitly, and give its ABCD matrix to show that a valid description still exists. [C05 | L5 | 2 marks]

P6. Cascade three two-ports of your choice using ABCD multiplication, then verify one result by directly analysing the complete loaded network. Comment on the effort of each route. [C05 | L5 | 2 marks]

- P7.** Determine the h parameters of a bipolar transistor small-signal model, label every unit, and compute the voltage gain into a stated load. [C05 | L4 | 2 marks]
- P8.** For one of your characterised networks, tabulate Z_{in} for five load values from short to open, and plot the result. Mark the two limiting values on the plot. [C05 | L4 | 1 marks]
- P9.** Compute Z_{out} for the same network with three different source impedances and identify the conjugate-matched load in each case. [C05 | L5 | 1 marks]
- P10.** Take a two-port containing at least one reactive element, compute its z parameters at three frequencies, and comment on how the terminated input impedance changes across them. [C05 | L5 | 1 marks]

Part B — Theory (3 marks)

- T1.** Explain why six parameter sets exist, organising your answer under the three distinct headings of existence, measurement practicality and algebraic convenience. [C05 | L5 | 1 marks]
- T2.** Explain why input impedance is not a property of a two-port alone, and relate the result to the maximum power transfer conclusions of Module 2. [C05 | L5 | 1 marks]
- T3.** Explain why reciprocity and symmetry are different claims, and construct one network that satisfies each condition without satisfying the other where that is possible. [C05 | L5 | 1 marks]

Submission guidelines

- Format: PDF, single file
- File name: <USN>_BEC303_Assignment5.pdf
- Channel: LMS upload
- Late policy: 10% per day, nothing after 5 days
- Every numerical answer must carry units and a stated reference direction.

Assessment rubric

Criterion	Weight	Excellent	Good	Satisfactory	Needs improvement
Parameter determination and terminations	6	All parameters derived from first principles with the termination stated and justified for each; units correct throughout including mixed-unit sets.	Parameters correct with terminations stated; minor unit omissions.	Parameters correct but terminations assumed rather than stated.	Wrong termination used, e.g. z_{11} computed with a short; units missing.
Conversion and existence reasoning	5	Conversions performed correctly with the divisor checked first; a non-existent set demonstrated explicitly and an alternative supplied.	Conversions correct with the determinant computed.	Conversions correct but no existence check.	Elementwise reciprocation used in place of matrix inversion.

Criterion	Weight	Excellent	Good	Satisfactory	Needs improvement
Cascading and equivalent circuits	4	Ordered ABCD products correct and verified against a direct analysis; π and T equivalents derived and independently confirmed.	Cascades correct with one verification.	Cascades computed without verification.	Matrices multiplied in the wrong order, or equivalents asserted without derivation.
Terminated performance and interpretation	5	Z_{in} , Z_{out} and gain computed correctly across the requested conditions, with limiting cases checked and the matching conclusion linked back to Module 2.	Correct values with at least one limiting-case check.	Values computed but limiting cases not checked.	z_{11} quoted as the terminated input impedance; gain quoted without stating the load.
Total	20				

Course outcome attainment

Course outcome	Marks carried
CO5	20
Total	20

Academic integrity

Individual submission. Identical submissions score zero for both parties.